Extracting Audience Feedback from comments using Opinion Mining and Text Summarization

Project report submitted to the Amrita Vishwa Vidyapeetham in partial fulfilment of the requirement for the Degree of

BACHELOR of TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

Submitted by

Gokul Krishnan G AM.EN.U4CSE19321

AMRITA SCHOOL OF COMPUTING

AMRITA VISHWA VIDYAPEETHAM
(Estd. U/S 3 of the UGC Act 1956)
AMRITAPURI CAMPUS

KOLLAM -690525

APRIL 2023

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

AMRITA VISHWA VIDYAPEETHAM
(Estd. U/S 3 of the UGC Act 1956)

Amritapuri Campus

Kollam -690525

BONAFIDE CERTIFICATE

This is to certify that the project report entitled **"Extracting Audience Feedback from user comments using Opinion Mining and Text Summarization"** submitted by **Gokul Krishnan G(AM.EN.U4CSE19321)**, in partial fulfillment of the requirements for the award of Degree of Bachelor of Technology in Computer Science and Engineering from Amrita Vishwa Vidyapeetham, is a bonafide record of the work carried out by them under my guidance and supervision at Amrita School of Computing, Amritapuri during Semester 8 of the academic year 2022-2023.

Dr. Divya Udayan J
Project Guide

Dr. Thushara M G
Project Coordinator

Dr. Prema Nedungadi
Chairperson
Dept. of Computer Science & Engineering

Reviewer

Place : Amritapuri
Date : 25 April 2023

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

AMRITA VISHWA VIDYAPEETHAM
(Estd. U/S 3 of the UGC Act 1956)

Amritapuri Campus

Kollam -690525

DECLARATION

I, **Gokul Krishnan G(AM.EN.U4CSE19321)** hereby declare that this project entitled **"Extracting Audience Feedback from user comments using Opinion Mining and Text Summarization"** is a record of the original work done by me under the guidance of **Dr. Divya Udayan J**, Dept. of Computer Science and Engineering, Amrita Vishwa Vidyapeetham, that this work has not formed the basis for any degree/diploma/associationship/fellowship or similar awards to any candidate in any university to the best of my knowledge.

Place : Amritapuri

Date : 25 April 2023

Signature of the student

Signature of the Project Guide

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Finally I extend my gratitude to anyone who contributed to this project both directly and indirectly.

Abstract

Social media comments sections provides the largest collection of information and insights about people's opinions, beliefs, and behaviours as they provide a platform for people to voice their opinions with minimal restrictions. This vast amount of data is very useful to understand the public's perception of certain topics and can be useful for businesses, service providers and content creators. This project proposes a way to extract audience feedback from comments section through opinion mining and text summarization, using state-of-the-art natural language processing techniques and fine-tuning large language models for specific downstream tasks. Initially, using a linear SVM, a spam filter is implemented to filter out spam comments. Opinion mining is done using multi label emotion recognition using fine tuned, modified BERT model. Keyword extraction from comments is also implemented by leveraging both textrank algorithm and tf-idf scores. For performing text summarization on the fetched comments both extractive and abstractive techniques were used, extractive summarization is implemented using BERT embeddings and K-means clustering, this limits the number of similar comments that makes it into the final results, abstractive summarization is performed on the results from the extractive summaries using a fine tuned t5 model. Finally a web application is designed by developing a flask API that returns the results of the project and a ReactJS website for displaying the results in a usable format.

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Chapter 1

Introduction

Comments from users on online forums is one of the most important ways a content creator or a product manufacturer can directly receive feedback from their main consumer base, be it praises or criticism or even insults. Having a general idea of the public's opinion on you or your service is very important in improving your product or service, understanding your target audience, streamlining marketing efforts and addressing complaints and grievances.

Opinion mining can provide insights into how people feel about certain topics which can help give a better understanding of what works in the current market or creative circle. Fine grained emotion recognition is very helpful in this regard as it helps detect a lot of different emotions in text which can be further analyzed to understand what the consumers are feeling about the particular product, service or topic of discussion.

Normally Text summarization is used to condense text while retaining alot of the core meaning, in the context of extracting feedback from comments section summarization can be used to condense and extract only insightful comments and re-write them in a uniform way, so this makes it easier

to read and understand insightful audience opinions without having to read through thousands of comments.

In this project I am implementing a pipeline to extract feedback from user comments on social media platforms, I will be focusing on youtube comments as youtube is one of the largest and most popular social media platforms and a lot of times youtube comments can be descriptive and insightful unlike other popular platforms like facebook or instagram. But the work done in this project can be extended to other platforms as well even on amazon product reviews.

The planned workflow includes fetching youtube comments through the youtube API, using machine learning algorithms to filter out spam comments and non-english comments, preprocessing and tokenizing the cleaned data, performing sentiment analysis on the filtered comments to understand the public reception of the content, perform text summarization using deep learning techniques to generate a feedback report of the comments that gives the users valuable insights into the public's opinions related to the content of the video.

Example use cases:

- A new movie is launching and the production has put out a trailer to youtube. By analyzing the youtube comments on the trailer video, we can get an idea of how excited the audience is for the movie or if they are disappointed, get a general idea of how well the movie could perform on the box office and plan the launch and marketing accordingly.
- Say a new product is being launched, for eg samsung is launching the new galaxy s23 ultra, they uploaded to youtube, a trailer video advertising their new device and they want to understand the public reception

of their device, they can use this project to get insights into the public opinion about their product, public's perception of their brand, know whether the public reception of their product is positive or negative and focus on marketing efforts in the right direction.

- The same can be applied to all popular videos on youtube, music videos, video game trailers, independent youtubers, vlogs etc..

Chapter 2

Problem Definition

Content creators and product manufacturers often struggle to get direct and constructive feedback from their audience and consumers, While online forums and social media platforms provide a convenient way to gather feedback, the sheer volume of comments and the presence of spam and non-constructive comments can make it difficult to extract meaningful insights. Additionally, manually reading and analyzing thousands of comments is a time-consuming and labor-intensive process that may not be feasible for many creators and manufacturers. This can lead to an unhealthy disconnect from audience that can lead to poor content, wrong marketing strategies, overall the product or service not being able able to impress potential customers and loosing business. Natural Language Processing techniques and Large language models can provide a cost-effective and efficient way to extract insights from large volumes of textual data. This project uses state-of-the-art NLP algorithms and large language models to develop a system that can perform fine grained emotion recognition on comments and also extract keywords and summarize comments, generally and as a whole and also about specific topics.

Chapter 3

Related Work

Earlier work related to each individual component in this project is summarized in this section.

3.0.1 Spam Filter

Spam filtering has been an area where vast amounts of research has been done, on almost all kinds of data, whether it be emails, messages, comments etc. Datasets have also been developed specifically for predicting youtube spam comments. Spam comments on youtube are not a huge problem anymore as youtube already has a good spam filter in place, So, with respect to this project spam filtering is not huge problem that needs to be addressed so a simple machine learning algorithm will suffice.

3.0.2 Multi label Emotion Recognition

The problem of sentiment analysis has been explored deeply and thoroughly in recent years, most of the work related to multi label emotion recognition has been based on the SemEval dataset that only has a limited emotion range.

New more detailed datasets have been released, like Google's GoEmotions dataset, that can be used for much fine grained emotion recognition and these datasets have not been explored properly. Current advancements in sentiment analysis or opinion mining includes using large language models like BERT, which are trained on a large corpus of text data and can be fine tuned for many downstream tasks like sentiment analysis, they provide state-of-the-art results on most major nlp tasks.

3.0.3 Keyword Extraction

Two popular methods for keyword extraction are using tf-idf and textrank algorithms, these are easier to implement and use as they are not computationally expensive and give decent performance, recently a new technique has been introduced that leverages both textrank and tf-idf scores to extract keywords from text.

3.0.4 Text Summarization

Text summarization is also area of natural language processing where a lot of reseach is being done, particularly using language models. Language models like Googles T5 model and BERT are have been used for both extractive and abstractive summarization techniques with state-of-the-art results. Summarization techniques combining both extractive and abstractive methods have been recent topic of research in text summarization tasks.

Research gap:

- Fine grained emotion recognition, to detect a vast range of emotions has been challenge, eventhough a lot of research has been done in this

field, one major problem is the difficulty to accurately identify sarcasm in comment.

- For this process of summarizing comments from a comment section, a major problem that has not been addressed yet is the problem of multiple similar comments that can make it into the results of abstractive summaries and make it less comprehensive.

Chapter 4

Requirements

The design of this project contains both hardware and software. The specifications are listed below.

4.1 Hardware

- **CPU**: Recommended (In order to run the api and website server): intel core-i7 8th generation and up or AMD equivalent
- **Memory**: At least 8GB of DDR5 RAM
- **GPU**: For training the models locally, I recommend using atleast an NVIDIA RTX series GPU with 8GB dedicated VRAM or an NVIDIA Tesla series GPU
- **Disk**: Minimum 5GB of free disk space
- High speed internet connection, atleast 20Mbps, in order fetch comments from youtube API at reasonable speed.

4.2 Software

- Python version 3.8 and up. (all required libraries and versions are mentioned in the requirements.txt file)
- Huggingface transformer library for loading datasets and using language models.
- Google Colab runtime or Kaggle notebook in order to access much capable hardware for free or cheap.
- NodeJs v19.8.1 and other javascript libraries added in package.json file.
- PyCharm or VS-Code code editor mainly for easy set up of a virtual environment and debugging features.

Chapter 5

Proposed System

This project consists of 5 independent modules and Sections 5.1 to 5.4 describes each module.

5.1 Spam Filter

Spam comments are not really a huge problem since youtube already removes most of the spam comments and is pretty efficient, but a dedicated spam filter is implemented just in case, Which makes this part not an important part of the project, so the spam filter is implemented using simple machine learning algorithms which gives satisfactory results while still running efficiently and fast enough that it doesn't become a bottleneck. In total 5 machine learning algorithms are implemented and their performances are compared to find the best one for this task.

5.1.1 Logistic Regression

Logistic regression is a type of supervised learning algorithm used for classification tasks in machine learning. The goal of logistic regression is to predict

the probability of an event occurring based on one or more input variables. The logistic regression algorithm works by fitting a logistic function (also known as a sigmoid function) to the input data. This function maps any real-valued number to a value between 0 and 1. The output of the logistic function is interpreted as the probability of the event occurring.

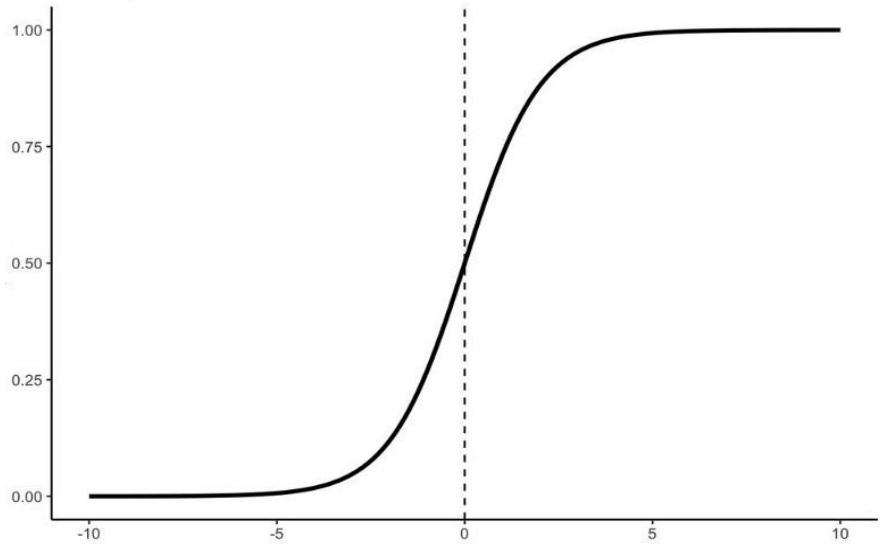

Figure 5.1: Sigmoid function

5.1.2 Linear SVM

Linear SVM, short for linear support vector machine, is a binary classification algorithm used to separate two classes of data. It works by finding the best possible line (hyperplane) that separates the two classes in a way that maximizes the margin between the classes. The margin is the distance between the hyperplane and the closest data points from each class. The hyperplane that has the maximum margin is the one that best separates the data points, and thus provides the most robust classification.

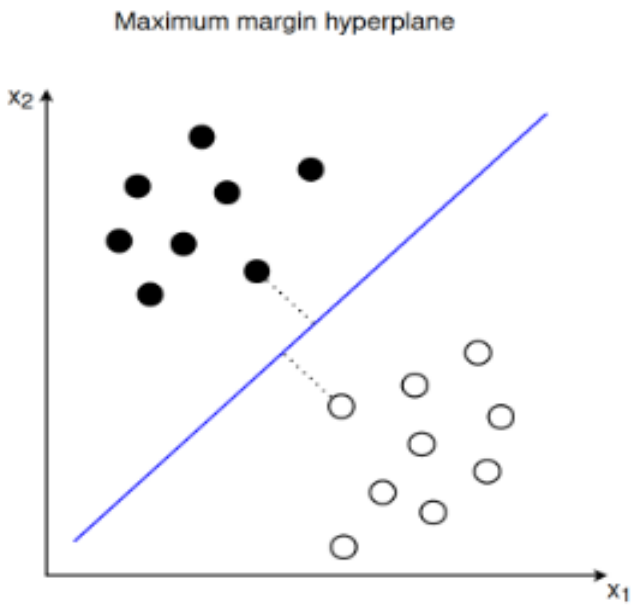

Figure 5.2: SVM on linearly seperable data

5.1.3 Decision Tree

A decision tree is a non-parametric, supervised learning algorithm that is commonly used in classification and regression tasks. It is a tree-like model in which each internal node represents a test on a feature, each branch represents the outcome of the test, and each leaf node represents a class label or a regression value. The decision tree algorithm recursively splits the data into subsets based on the feature that maximally separates the data into homogeneous classes or reduces the variance in regression tasks. The splitting criterion is chosen to optimize a certain metric such as information gain, gain ratio, or Gini impurity.

5.1.4 Random Forest Classifier

Random forest is a popular machine learning algorithm that is used for both classification and regression tasks. It is an ensemble method that combines multiple decision trees to form a "forest". Each decision tree in the forest

is trained on a different subset of the data, and at each node of the tree, a random subset of the features is considered for splitting. To make a prediction using a random forest, each decision tree in the forest makes a prediction, and the final prediction is determined by taking a majority vote of the individual tree predictions (in the case of classification) or averaging the individual tree predictions (in the case of regression).

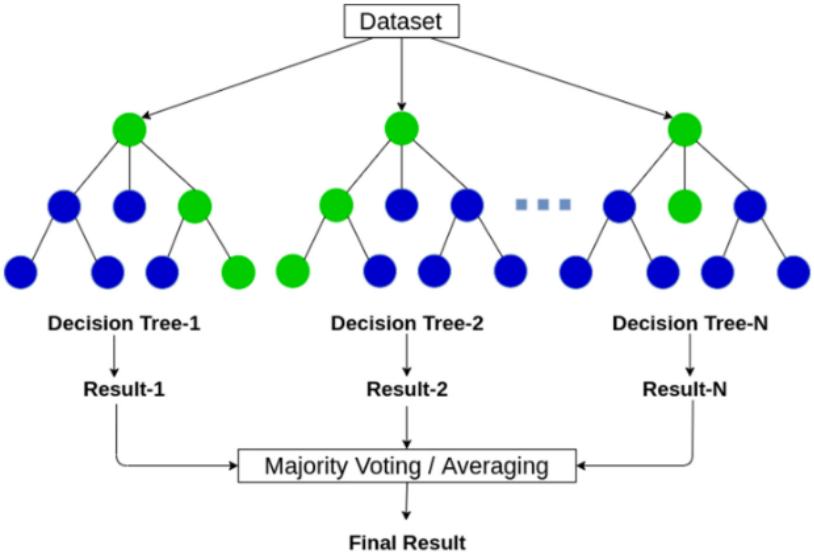

Figure 5.3: Random Forest Classifier

5.1.5 Naive Bayes

Naive Bayes is a probabilistic classification algorithm that is based on Bayes' theorem. It assumes that the presence of a particular feature in a class is independent of the presence of other features. This is known as the "naive" assumption, which simplifies the calculations and makes the algorithm computationally efficient. The algorithm uses the probability of each feature to calculate the probability of a class. It then assigns the class with the highest probability to the input data. The probability of a class given the input

data is calculated using Bayes' theorem, which states that the probability of a hypothesis (in this case, a class) given the data is proportional to the probability of the data given the hypothesis times the prior probability of the hypothesis.

5.2 Multi Label Emotion Recognition

The comments fetched from the youtube API are preprocessed to perform fine grained emotion recognition. In this project, a custom model made by modifying BERT language model with a Bidirectional LSTM layer is implemented and fine tuned for the task of multi label emotion recognition and is compared with a base model implemented using just a BERT uncased model. The final result of this module is the percentage of comments in the corpus that shows each of the 27 different emotions that model was trained on.

5.2.1 BERT

BERT (Bidirectional Encoder Representations from Transformers) is a pre-trained language model developed by Google AI Language. It uses a transformer architecture that allows it to learn contextual relations between words in a text corpus.

Unlike previous language models that are trained to predict the next word in a sentence based on the previous words, BERT uses a technique called "masked language modeling". In this technique, some of the words in the input sentence are randomly masked and the model is trained to predict the masked words based on the context of the surrounding words. This allows the model to understand the relationships between all the words in the sen-

tence.

BERT also uses another technique called "next sentence prediction". In this technique, the model is trained to predict whether a sentence follows another sentence in a document. This allows the model to understand the context and flow of text in a document.

The pre-trained BERT model can be fine-tuned for specific natural language processing (NLP) tasks such as sentiment analysis, named entity recognition, and question answering. Fine-tuning involves training the model on a smaller task-specific dataset, allowing it to learn task-specific features.

Overall, BERT is considered one of the most powerful and versatile pre-trained language models for NLP tasks.

```
__________________________________________________________________________________________________
 Layer (type)                   Output Shape         Param #     Connected to
==================================================================================================
 attention_mask (InputLayer)    [(None, 33)]         0           []

 input_ids (InputLayer)         [(None, 33)]         0           []

 token_ids (InputLayer)         [(None, 33)]         0           []

 bert (TFBertMainLayer)         TFBaseModelOutputWi  109482240   ['attention_mask[0][0]',
                                thPoolingAndCrossAt                 'input_ids[0][0]',
                                tentions(last_hidde                 'token_ids[0][0]']
                                n_state=(None, 33,
                                768),
                                 pooler_output=(Non
                                e, 768),
                                 past_key_values=No
                                ne, hidden_states=N
                                one, attentions=Non
                                e, cross_attentions
                                =None)

 pooled_output (Dropout)        (None, 768)          0           ['bert[0][1]']

 emotion (Dense)                (None, 27)           20763       ['pooled_output[0][0]']

==================================================================================================
Total params: 109,503,003
Trainable params: 109,503,003
Non-trainable params: 0
__________________________________________________________________________________________________
```

Figure 5.4: Bert Classification model

5.2.2 BiLSTM - BERT classification model

BERT is a powerful pre-trained language model that uses a transformer architecture to learn contextual relations between words in a text corpus. How-

ever, BERT only processes input text in one direction, from left to right or from right to left, depending on the training mode.

Adding a bidirectional LSTM layer on top of BERT allows the model to take into account both the past and future context of a word in a sentence. The LSTM layer can capture sequential information and contextual relationships between words in a more comprehensive way than BERT alone, making it particularly useful for tasks such as named entity recognition and sentiment analysis.

Layer (type)	Output Shape	Param #	Connected to
attention_mask (InputLayer)	[(None, 48)]	0	[]
input_ids (InputLayer)	[(None, 48)]	0	[]
token_ids (InputLayer)	[(None, 48)]	0	[]
tf_bert_model_1 (TFBertModel)	TFBaseModelOutputWithPoolingAndCrossAttentions(last_hidden_state=(None, 48, 768), pooler_output=(None, 768), past_key_values=None, hidden_states=None, attentions=None, cross_attentions=None)	109482240	['attention_mask[0][0]', 'input_ids[0][0]', 'token_ids[0][0]']
bidirectional (Bidirectional)	(None, 128)	426496	['tf_bert_model_1[0][0]']
dropout_74 (Dropout)	(None, 128)	0	['bidirectional[0][0]']
output (Dense)	(None, 27)	3483	['dropout_74[0][0]']

Total params: 109,912,219
Trainable params: 109,912,219
Non-trainable params: 0

Figure 5.5: BiLSTM-BERT Classification model

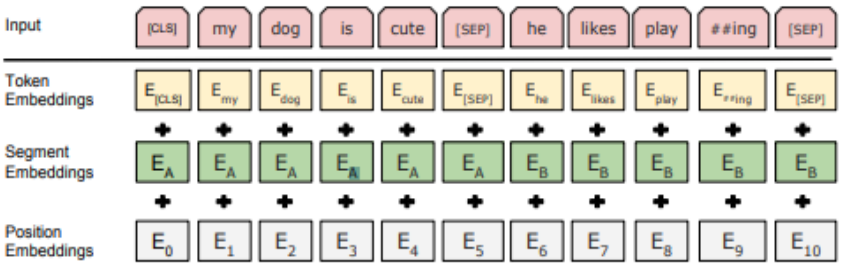

Figure 5.6: BERT input representation

5.3 Keyword Extraction

Extracting keywords like proper nouns and adjectives from a large collection of comments will give an idea of what viewers are saying about named entities like people, places and organisations.
Two popular methods for doing keyword extraction are using TF-IDF scores and textrank algorithm and both methods has its own advantages.
TF-IDF method is better able to find unique words that occur less frequently in the document while textrank also takes into account the relationship between words, which makes it more efficient.
In this project, to make the most of both techniques, the scores given to the words by both techniques will be added and for better results only proper nouns and adjectives will be considered as keywords.

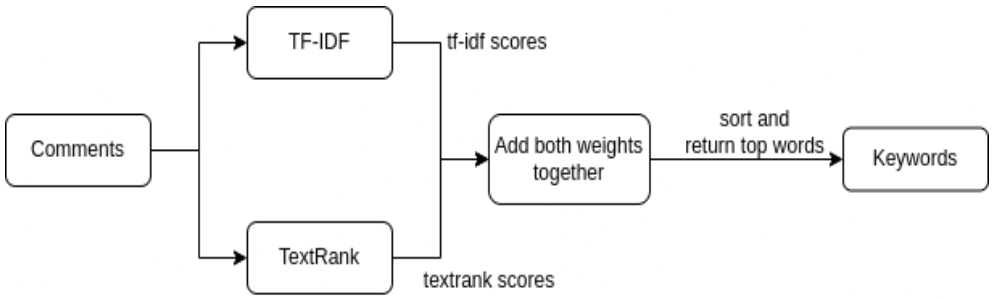

Figure 5.7: Keyword extraction using Tf-Idf and Textrank

5.4 Text Summarization

Text summarization in this particular context is a little complicated, mainly because the text to be summarized is a large collection of comments where some of them are similar

5.4.1 Extractive Summarization

In order to limit the number of similar comments from the corpus, extractive summarization is implemented using BERT embeddings and Kmeans clustering. The individual comments are converted to vector embeddings using BERT language model. Kmeans clustering is used to cluster together vector embeddings that are similar to each other and only sentences from the centroid are selected to get extractive summarization results.

5.4.2 Abstractive Summarization

A T5-small model is fine tuned on a small subset of the reddit comments dataset to implement abstractive summarization part. Abstractive summarization is used to further shorten the text and bring some uniformity to the final results.

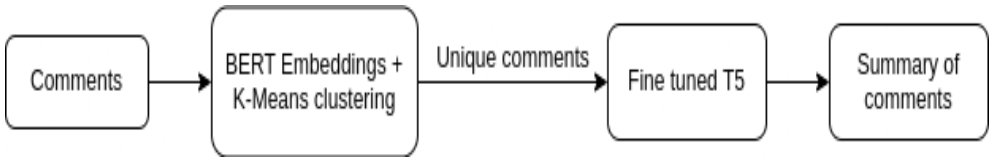

Figure 5.8: Text summarization combining extractive and abstractive techniques

5.5 Web Application

A flask API is implemented with 5 endpoints that return the outputs of the above mentioned models. The ReactJs frontend takes a youtube video link as input extracts the videoid from the link, uses that to fetch data from the API and displays it.

5.6 Final System

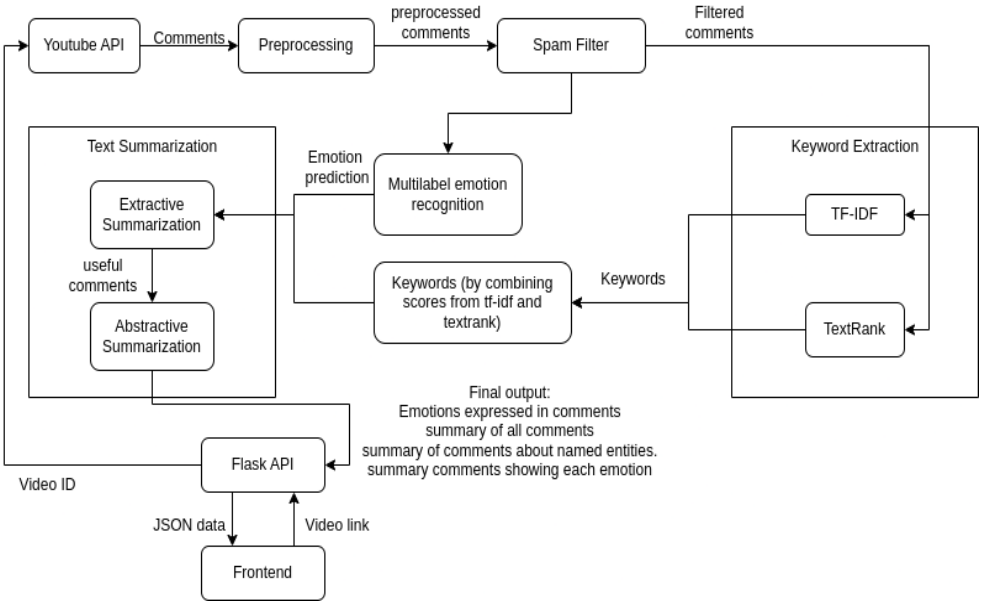

Figure 5.9: Final system design

Chapter 6

Result and Analysis

6.1 Spam Comments Filter

5 different machine learning algorithms were tested for this module and it is observed that linear SVM is best suited for this task both in terms of execution speed and quality of predictions.
LSVM gave an accuracy of 91% and an F1 score of 0.912, which is more than enough for this project, Table 6.1 is a comparison of results produced by all tested models.

Algorithm	**Accuracy**	**Precision**	**Recall**	**F1 Score**
Linear SVM	**0.91**	0.864	0.967	**0.912**
Logistic Regression	0.90	0.860	0.952	0.90
Decision Tree	0.88	0.953	0.847	0.897
Naive Bayes	0.70	0.582	0.789	0.67
Random Forest	0.89	0.950	0.856	0.90

Table 6.1: Comparing The Performance of Five Algorithms for Spam Filter

6.2 Multi label Emotion Recognition

For multi label emotion recognition two models were tested, a base model which is a fine tuned BERT model with a dropout layer and a novel model which is a combination of BERT and BiLSTM both capable of detecting 27 different emotions.

6.2.1 BERT model

Results for multilabel emotion recogniton using Bert model.
BERT model gave a F1 score of 0.46, from Figure 6.1 it is clear that the validation loss keeps increasing significanlty while train loss is reducing, this is a clear sign of overfitting, as a solution the training process is stopped after 4 epochs.

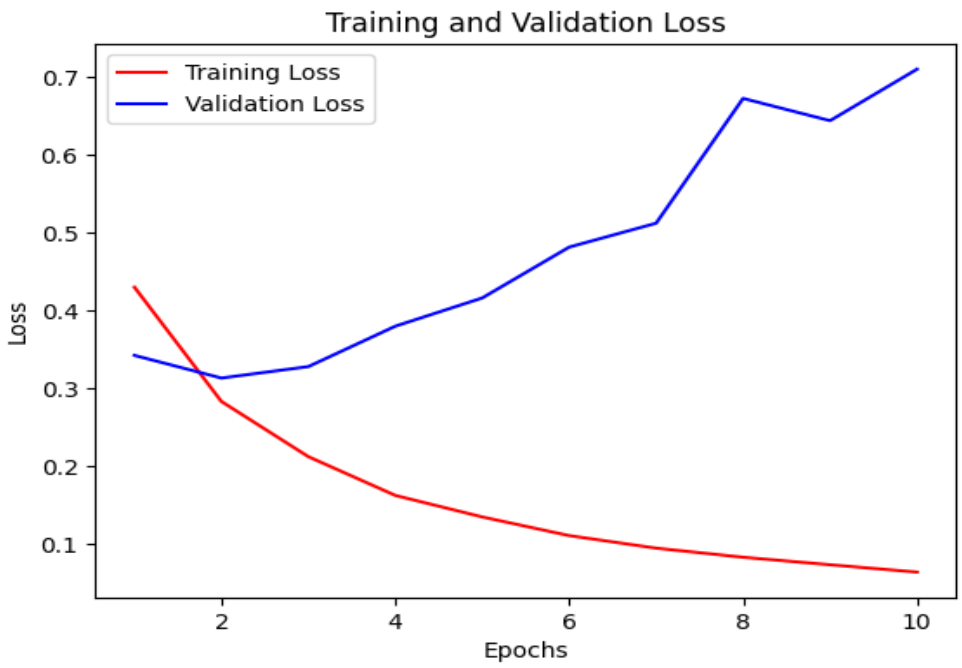

Figure 6.1: Train, Validation loss at each epoch for BERT

The model assigns a certain prediction confidence value to each label and a certain threshold value is used to get predictions using the model, Fig 6.2 shows variation in F1 score for each possible threshold value.

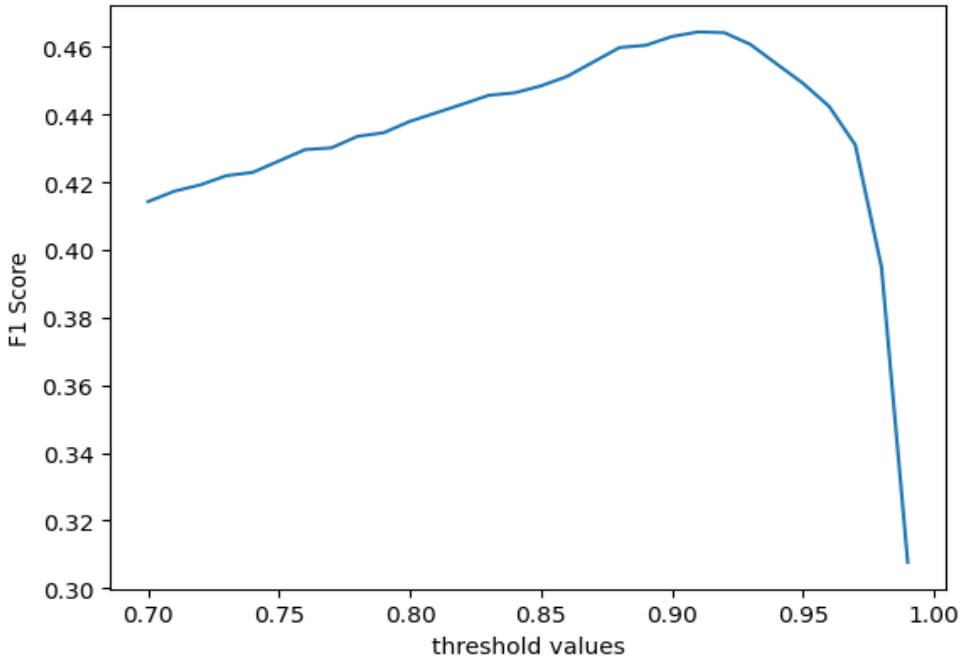

Figure 6.2: Threshold vs F1-Score for BERT

Table 6.4 shows the final prediction results of the model for each emotion type.

6.2.2 BiLSTM-BERT model

Results for multilabel emotion recogniton using Bilstm-Bert model.

BiLSTM-BERT model gave a slightly higher F1 score, 0.48, than BERT model, which is not much of an improvement on the test dataset, but it was to generalize better to unseen data and make more more coherent predictions on new unseen data. The problem of the model overfitting after 4 epochs is also there for BiLSTM-BERT model too.

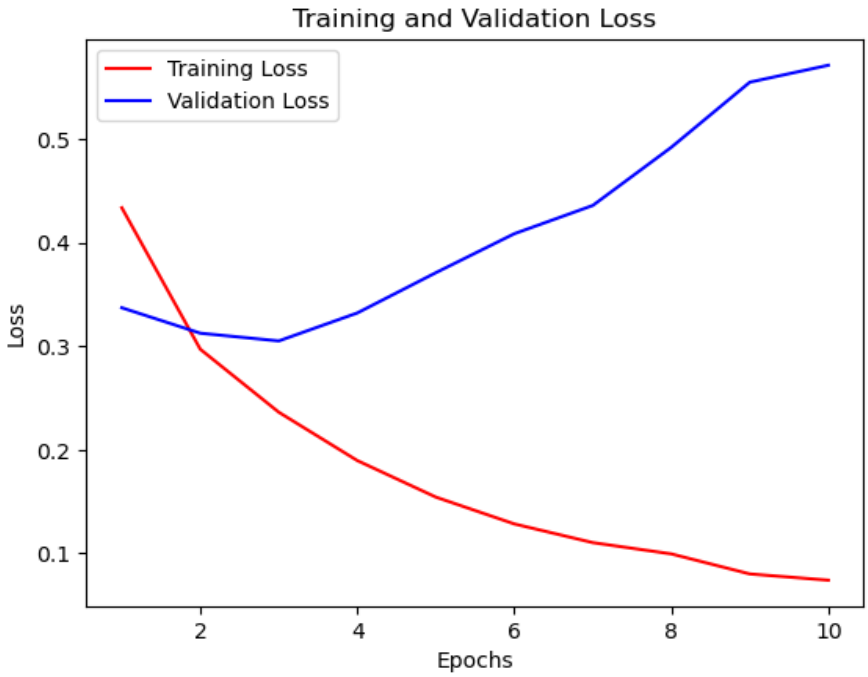

Figure 6.3: Train, Validation loss at each epoch for BiLSTM-BERT

Similar to BERT model this model also assigns a certain prediction confidence value to each label and a certain threshold value is used to get predictions using the model, Fig 6.5 shows variation in F1 score for each possible threshold value.

Model	**Precision**	**Recall**	**F1 Score**
BERT	0.41	0.60	0.46
BiLSTM-BERT	0.47	0.52	0.48

Table 6.2: Comparing Performance of BERT and BiLSTM-BERT for fine grained emotion recognition

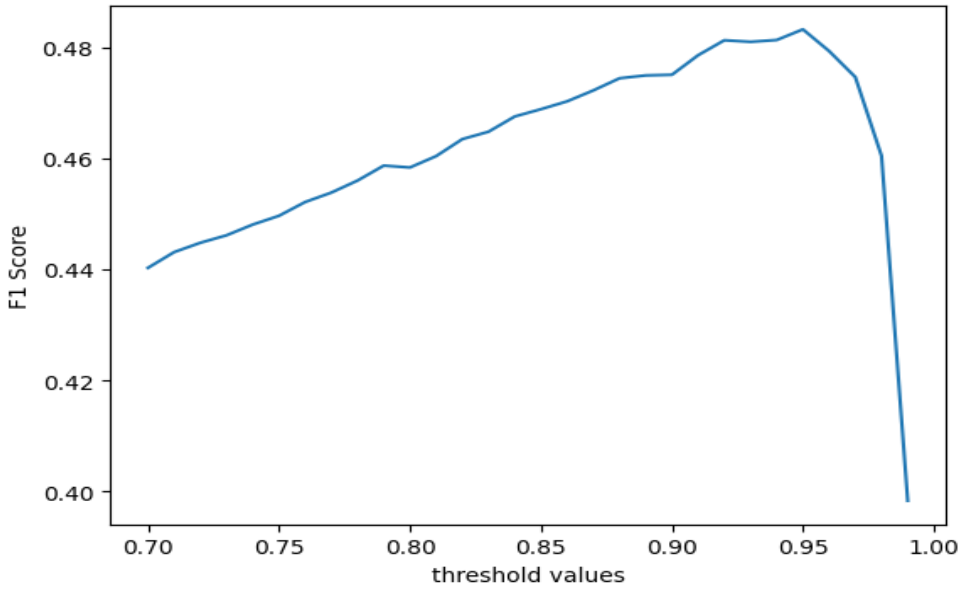

Figure 6.4: Threshold vs F1-Score for BERT

	Text	Emotions
0	It's like watching your parents wedding video ...	[]
1	The fact they are using a Snapdragon chip for ...	[admiration, approval]
2	Charlie being a Walmart shopper makes him more...	[admiration]
3	This brings tears to my eyes everytime I watch...	[disappointment, love, sadness]
4	I've literally played this 100 times already a...	[pride, realization]
5	Not only is the cooking worse, but the persona...	[annoyance, disgust]

Figure 6.5: Examples of BiLSTM-BERT model on real data from comment sections

6.3 Keyword Extraction

The following sentences are an example of a corpus of comments fetched from youtube API and concatenated together

These comments are from a video talking about the recent arrest of Justin Roiland, who is an american voice actor and producer best as known as the co-creator of Rick and Morty.

"justin roiland use to stand at a window in the office with binoculars and watch pedestrians walk around for inspiration for characters. because he

thought that nobody could see him through the window, and eventually found out people could see them, and had to stop. i found that fact really funny a couple weeks ago now i know what he does when no one is watching. honestly, i dont know why people think that rick morty wouldnt last without roiland. i never got the impression that he put a lot of muscle behind his writing contributions. to me, it was a dan harmon production featuring roiland doing voice acting. the sexual jokes around summer really hit different now. the show cannot survive without him he literally voices half the cast - justin roiland quote as described by sam hydedy when i get home from doing cartoons justin is not the mastermind behind rnm. its dan harmon (and the writers) who make the creative work. justin did the voices of rick and morty but harmon can also do the va if he wants to do it he can. i didnt know what it was going to be, but signing a show for 8 seasons in advance is wild. i knnneeewww something was going to pop off. too much happens over the span of a decade. particularly when youre dealing with artistic/creative peoplealways some lose screws somewhere. (i say as a creative person). it feels sorta disrespectful to all the talented people that work of rick and morty to say it was carried by justin theres nothing shwifty about domestic abuse" is my new favorite sentence theres nothing shwifty about domestic violence. thank you, charlie. your wisdom enriches us all. i dont think its impossible that rick and morty could survive this, i imagine there are plenty of talented impressionists ready to come on the show, and as for the writing i think it could be good to have a change of pace to say the least"

Output of Keyword extraction by combining tf-idf and textrank scores:

{'justin': 2.265545138888889,

'rick': 1.6293576388888888,

'dan': 1.11,

'harmon': 1.165,

'morty': 1.1393090277777778,

'va': 0.9743090277777777,

'-justin': 0.7775833333333333,

'sam': 0.8325833333333333,

'sorta': 0.7732604166666667,

'roiland': 0.7080520833333332,

'charlie': 0.20500000000000002}

From the example ii is clear that the algorithm was able to extract most of the named entities and keywords from the input text.

6.4 Text Summarization

Model	Test Loss	Rouge1	Rouge2	RougeL
T5-small	3.471	16.8554	3.70	13.18

Table 6.3: Comparing Performance of BERT and BiLSTM-BERT for fine grained emotion recognition

The following is a collection of a few comments from the new Avatar movie trailer:

"i was one of the first creature designers who worked on this film (worked on the flying fish/crocodile creatures, the whale creatures, and more with a bunch of other insanely talented people), and i just wanted to say thank you to all of you, the fans. some of us who worked on the movies do take the time to read through the comments on the trailers, and seeing the overwhelmingly positive response from you all so far has made the years of work that have gone into these films worth it for all of us. truly, thank you. i'll see you in the theater in december! my daughter was 6 years old when i took her to

the cinema to watch the first avatar movie. now she is taking me to watch the second part, how time flies by ...i just wanna give the biggest round of applause to the director, actors, and the whole team. truly a masterpiece just seen the movie today in 3d. this is not just the movie of the year people. its the movie of the decade. incredible. the best visuals ive ever seen in a film. 10/10.for those who are contemplating on watching this movie at the theater....do not hesitate and go watch it, despite the media saying it didn't live up to the hype and all those other nonsense bs etc... if you were impressed with the 1st one, be prepared to be blown away with this sequel. true masterpiece....10/10. it's james cameron we're talking about, he makes movies for the ages... this movie is of epic proportions and it delivered on every level. you need to watch it at the theater to get the full experience, esp with the 3d. just wow this movie was 10 out of 10. 100% will watch againrest in peace for the people who adore the first avatar aswell as composer of the first movie, james horner who didn't get the chance to see this beautiful sequel that got to be made this wasn't just a movie, it was an experience. kudos to james cameron n the team to bring us these stunning visuals in imax 3di just saw it even though i didn't watch the first part before but my brother insisted me to see it and i can't express how glad i'm that i watched it. it was just magical. i came back statisfied from the movie theatre after a very long time. and guess what i did after coming back to home, i watched the first part and now i'm officially an avatar fanafter getting out of the theater, to all of the avatar fans who couldn't be here, this movie shines after 13 years. bravo to james cameron and everyone else! "i see you". it was unlike anything i've ever seen after returning to pandora! i just wanted to stay in that world. pure escapism by james cameron. by the way "never doubt james cameron"i watched it at theatre twice, a week apart.

no words describe the experience. just waw. i was in high school when the first movie came out, its absolutely magical and beyond exceptional to be back to pandora after over a decade. cant wait for avatar 34 and beyond. thank you james cameron.this movie's positive environmental theme is admirable, the beauty of pandora is overwhelming and it's creatures utterly captivating. in my 40 plus years of cinema going, this is the best experience i've witnessed. many thanks to james cameroon and his amazing team for this ground breaking movie.we need more trailers like this. it doesn't give everything away about its plot, it doesn't try to be action-y or flashy, it just lets you be fascinated.the movie in all its forms was absolutely phenomenal. gorgeous, stunning visuals, along with amazingly beautiful story. i absolutely fell in love with avatar again after 13 years from its original release. thank you for helping make this masterpiece. and hope to see more of it in the near future. 3 hours and 15 minutes, when i tell you this film did not feel like it was that runtime. truly a masterpiece. so excited for avatar 3just saw it and was genuinely blown away. probably the best cinematic experience i've ever had. better than the first in every way from story to acting, a perfect sequel and another technological triumph. seeing it again tomorrow as too much to take in.the storyline, connection to mythology spirituality is the soul of the movie. vfx are jaw dropping. a must go movie. worth waiting a decade!i just watched avatar: way of the water and i can truly say that this is one of the most stunning gorgeous and visually incredible movies i have ever seen. james cameron and all the actors are truly incredible and have incredible talents. how they are able to create this stunning movie is beyond me. far beyond a 10/10 movie. must see!!anyone else get goosebumps during the crescendo at 1:31 as that tulkun comes out of the water and sends tuk, tsireya, and lo'ak in the air?the amount of perseverance, dedication, talent,

patience, technology, devotion, resilience, passion and so much more speaks itself in such extraordinary volumes for this masterpiece that james cameron and the entire production have created over the years and i don't think i have words to even describe my appreciation and gratitude for them to bless us with this wonderful piece of art. thank you all so much. not only these trailers but the qualities that everyone have invested in this film has made me to actually genuinely cry and i don't even do that. this was certainly a very rare and if i have to say a unique moment in my life to be gifted with such a blessing. thank you so very much. you don't know how much your devotion to this means to me. thank you so very much. i wish you all of you the best of success for this film and in your lives as well. take care of yourselves and stay safe and stay happy. all of you are the best"

Output produced by fine tuned t5-small model:

"i was one of the first creature designers who worked on this film. james cameron and all the actors are truly incredible and have incredible talents - and they are able to create this stunning movie is beyond me. the film's positive environmental theme is admirable, the beauty of pandora is overwhelming and its creatures utterly captivating."

It is clear that the model was able to extract useful comments about the movie and some people behind its production.

Emotion	**Precision**	**Recall**	**F1 Score**
admiration	0.72	0.59	0.65
amusement	0.76	0.81	0.78
anger	0.38	0.56	0.45
annoyance	0.35	0.32	0.33
approval	0.53	0.26	0.35
caring	0.33	0.47	0.39
confusion	0.29	0.55	0.38
curiosity	0.49	0.57	0.53
desire	0.47	0.55	0.51
disappointment	0.31	0.34	0.33
disapproval	0.36	0.41	0.38
disgust	0.28	0.71	0.40
embarrassment	0.21	0.62	0.32
excitement	0.28	0.53	0.37
fear	0.42	0.83	0.56
gratitude	0.94	0.88	0.91
grief	0.21	1.00	0.34
joy	0.44	0.65	0.52
love	0.72	0.87	0.79
nervousness	0.18	0.43	0.26
optimism	0.53	0.49	0.51
pride	0.21	0.44	0.29
realization	0.23	0.31	0.26
relief	0.11	0.73	0.19
remorse	0.53	0.89	0.67
sadness	0.50	0.67	0.58
surprise	0.42	0.65	0.51
Macro-Average	**0.41**	**0.60**	**0.46**

Table 6.4: Precision, Recall, F1-Score for BERT model

Emotion	**Precision**	**Recall**	**F1 Score**
admiration	0.67	0.58	0.62
amusement	0.79	0.75	0.77
anger	0.51	0.39	0.44
annoyance	0.45	0.20	0.27
approval	0.46	0.27	0.34
caring	0.43	0.30	0.35
confusion	0.42	0.48	0.45
curiosity	0.49	0.61	0.54
desire	0.48	0.52	0.50
disappointment	0.25	0.36	0.29
disapproval	0.39	0.30	0.34
disgust	0.50	0.49	0.49
embarrassment	0.31	0.43	0.36
excitement	0.43	0.42	0.42
fear	0.55	0.76	0.64
gratitude	0.93	0.89	0.91
grief	0.27	0.67	0.38
joy	0.55	0.63	0.58
love	0.79	0.77	0.78
nervousness	0.21	0.48	0.29
optimism	0.54	0.51	0.52
pride	0.30	0.50	0.37
realization	0.25	0.30	0.27
relief	0.29	0.64	0.40
remorse	0.53	0.80	0.64
sadness	0.47	0.63	0.54
surprise	0.57	0.49	0.53
Macro-Average	**0.47**	**0.52**	**0.48**

Table 6.5: Precision, Recall, F1-Score for Bil-BERT model

Chapter 7

Conclusion

In this project i have a designed, implemented and tested a method to extract public feedback from the comments sections of youtube videos.
The system initially fetches a specified number of comments from youtube API filters out spam comments, performs fine grained multi label emotion recognition on the comments returns emotions displayed in each comment along with the percentage of comments in the corpus displaying each of the 27 different emotions the model is trained to detect. The model also displayed an F1 score of 0.48 on test dataset.
Keywords from the text, like named entities, Organizations etc.. are also extracted, in addition to this the user can provide their own keywords which they are interested in getting more information about. Finally the comments are summarized and results are returned to the users, in addition to summarizing the entire comment section users can also fetch summaries about comments mentioning specific entities, keywords and also comments showing specific emotions. Finally a website is developed to process and display the results mentioned above. Overall this project is very useful in understanding the audiences perception about certain topics, products, people etc.. the

information produced by this system can be used to supplement marketing efforts, align content creation to fit the audience and many other applications where consumer feedback is valuable.

Major contributions of this Project:

- A new model based on BERT language model and bidirectional LSTM for fine grained emotion recognition, that can detect multiple emotions expressed in a piece of text and can recognize upto 27 different emotions.
- Implemented a more efficient text summarization method for this specific task of summarizing comments by combining both extractive and abstractive text summarization techniques to limit multiple similar comments making into final result.
- Designed and developed an interactive Web application, using Flask, ReactJS and three.Js.

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Appendix A

Source code

A.1 Summary of Datasets

A.1.1 Youtube Spam Collection v1

This is a public dataset of comments collected for spam research. It has five datasets composed by 1,956 real messages extracted from five videos that were among the 10 most viewed on the collection period.

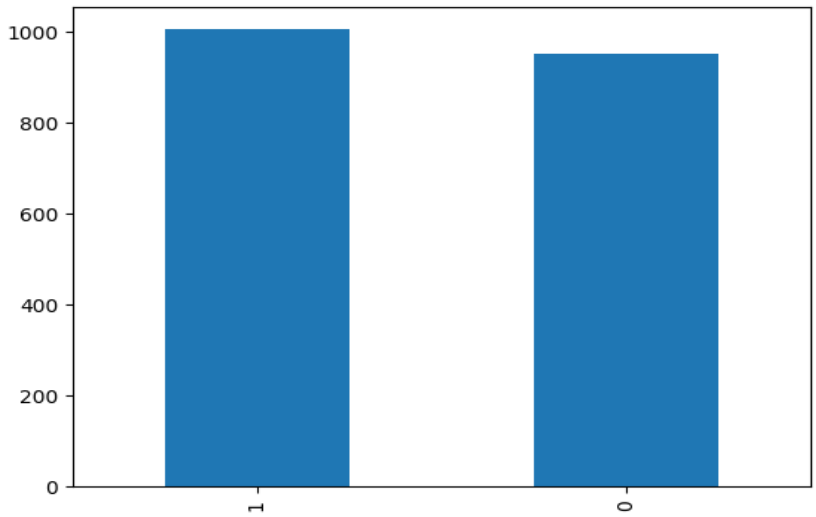

Figure A.1: Data distribution

A.1.2 GoEmotions

GoEmotions dataset is a human annotated collection 58k Reddit comments extracted from English language subreddits and labelled with 27 emotion categories. In contrast to the basic six emotions, which include only one positive emotion (joy), this taxonomy includes 12 positive, 11 negative, 4 ambiguous emotion categories and 1 "neutral", making it widely suitable for conversation understanding tasks that require a subtle differentiation between emotion expressions.

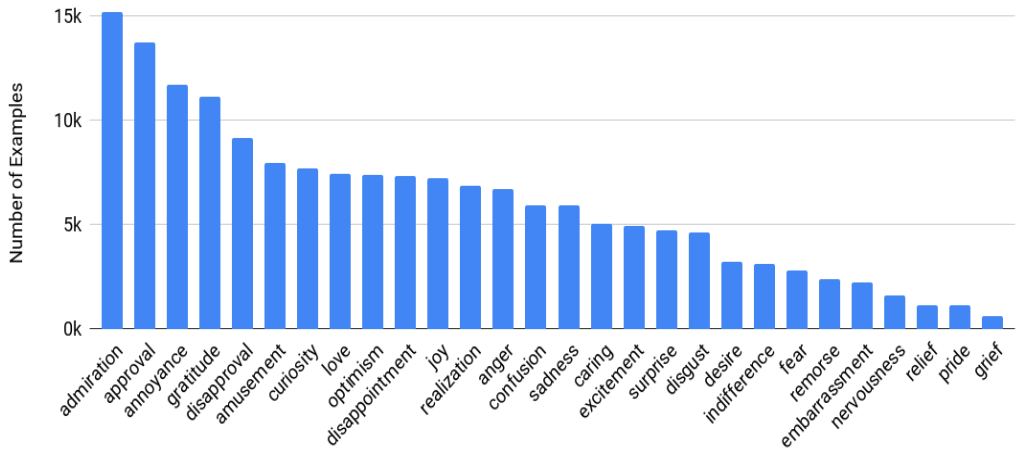

Figure A.2: Distribution of emotions in the dataset

A.1.3 Reddit Comments Summary

This project uses a small subset of the reddit comments dataset from huggingface that contains text, which are reddit comments, and summaries of the comments. The dataset contains 26938 samples for train, 5773 samples for test, 5772 samples for test.

Figure A.3 shows some samples from train split.

	text	summary
21171	When I was six or so, my family went to Six Fl...	I had my head slammed against the side of a ro...
32709	I had this negative attitude built-in to my la...	Goes for guilds too, don't settle on a negativ...
11753	My uncle made out with a transvestite and felt...	General Shenannies
16186	No, vanilla test means 2*cost+1 like Yeti and ...	he brings too much value in combination with h...
24870	I used to attend church with a elderly man sur...	I know a guy whose last name means toilet, a g...
...	...	...
6265	Same here. My wife makes me. Basically, here...	Get a good wife.
11284	Roku 2 XS and Plex media server. Why reinvent ...	Roku 2 SlapChop
38158	Replies to genuine reactions to a username don...	stewarted started it.
860	Hello people. I do a lot of coding in [SuperCo...	Is there any way to change the default ruler s...
15795	Pretentious redditor is pretentious. \n You've...	Fuck you and your immediate "filthy Americans"...

26938 rows × 2 columns

Figure A.3: Sample entries from reddit comments dataset

A.2 Data Preprocessing

Preprocessing is different for different modules in this project. Some parts like text summarization doesn't need any preprocessing as all information should be passed into the model for generating coherent output.

A.2.1 Spam filter

For spam filter all the basic preprocessing steps like stopwords removal, removing links hashtags and user mentions, removing digits and punctuations, stemming and lemmatizing words and vectorizing the sentences using CountVectorizer.

```
[ ] def de_repeat(text):                                                    #removing repeated characters in wor
      pattern = re.compile(r"(.)\1{2,}")
      return pattern.sub(r"\1\1", text)
```

```
[ ] def remove_html(raw_comment):
      raw_comment = re.sub(r'<br>', " ", raw_comment)
      raw_comment = re.sub(r'<b>', " ", raw_comment)
      raw_comment = re.sub(r'</b>', " ", raw_comment)
      raw_comment = html.unescape(raw_comment)
      raw_comment = re.sub(r'\n', "", raw_comment)
      raw_comment = re.sub(r'\t', "", raw_comment)
      raw_comment = re.sub(r'<[^<]+?>', '', raw_comment)
      return raw_comment
```

```
[ ] df['Comment'] = df['Comment'].apply(remove_html)
    df['Comment'] = df['Comment'].str.replace("@[A-Za-z0-9_]+", "")                  #The text in the datasets are
    df['Comment'] = df['Comment'].str.replace("#[A-Za-z0-9_]+", "")                  #These have to be removed
    df['Comment'] = df['Comment'].str.replace("^https?:\/\/.*[\r\n]*", "")
    df['Comment'] = df['Comment'].apply(lambda x: " ".join(x.lower() for x in x.split())) #changing text to lowercase
    df['Comment'] = df['Comment'].str.replace('[^\w\s]',' ')                         #removing punctuation and digi
    df['Comment'] = df['Comment'].str.replace('[0-9]', ' ')
    df['Comment'] = df['Comment'].apply(lambda x: " ".join(x for x in x.split() if x not in stop))        #removing stopw
    df['Comment'] = df['Comment'].apply(lambda x: " ".join([Word(word).lemmatize() for word in x.split()]))       #lemma
    df['Comment'] = df['Comment'].apply(lambda x: " ".join(de_repeat(x) for x in x.split()))
```

Figure A.4: Preprocessing data for spam filter

A.2.2 Multi label Emotion Recognition

In addition to the preprocessing steps for spam filter,for emotion recognition some more steps are done like removing emojis, expanding word contractions, expanding certain abbreviations and slang words.

```
def preprocess_corpus(x):

    x = re.sub( r'([a-zA-Z])([,;.!?])', r'\1 \2', x)
    x = re.sub( r'([,;.!?])([a-zA-Z])', r'\1 \2', x)

    x = emoji.demojize(x)
    x = contractions.fix(x)
    x = x.lower()

    x = re.sub(r"lmao", "laughing my ass off", x)
    x = re.sub(r"amirite", "am i right", x)
    x = re.sub(r"\b(tho)\b", "though", x)
    x = re.sub(r"\b(ikr)\b", "i know right", x)
    x = re.sub(r"\b(ya|u)\b", "you", x)
    x = re.sub(r"\b(eu)\b", "europe", x)
    x = re.sub(r"\b(da)\b", "the", x)
    x = re.sub(r"\b(dat)\b", "that", x)
    x = re.sub(r"\b(dats)\b", "that is", x)
    x = re.sub(r"\b(cuz)\b", "because", x)
    x = re.sub(r"\b(fkn)\b", "fucking", x)
    x = re.sub(r"\b(tbh)\b", "to be honest", x)
    x = re.sub(r"\b(tbf)\b", "to be fair", x)
    x = re.sub(r"faux pas", "mistake", x)
    x = re.sub(r"\b(btw)\b", "by the way", x)
    x = re.sub(r"\b(bs)\b", "bullshit", x)
    x = re.sub(r"\b(kinda)\b", "kind of", x)
    x = re.sub(r"\b(bruh)\b", "bro", x)
    x = re.sub(r"\b(w/e)\b", "whatever", x)
    x = re.sub(r"\b(w/)\b", "with", x)
    x = re.sub(r"\b(w/o)\b", "without", x)
    x = re.sub(r"\b(doj)\b", "department of justice", x)

    x = re.sub(r"\b(j+e{2,}z+e*)\b", "jeez", x)
    x = re.sub(r"\b(co+l+)\b", "cool", x)
    x = re.sub(r"\b(g+o+a+l+)\b", "goal", x)
    x = re.sub(r"\b(s+h+i+t+)\b", "shit", x)
    x = re.sub(r"\b(o+m+g+)\b", "omg", x)
    x = re.sub(r"\b(w+t+f+)\b", "wtf", x)
    x = re.sub(r"\b(w+h+a+t+)\b", "what", x)
    x = re.sub(r"\b(y+e+y+|y+a+y+|y+e+a+h+)\b", "yeah", x)
```

Figure A.5: Preprocessing data for Emotion Recognition model

A.2.3 Keyword extraction

For keyword extraction main preprocessing steps are to just get proper nouns and adjectives from the text using spacy library.

A.2.4 Text Summarization

For text summarization only preprocessing step needed is to split the text into senetences and add newline character as seperator for the generating embeddings using BERT model.

A.3 Training

Model training is only involved for the following sections

A.3.1 Spam Filter

Training Linear SVM model

```
model = SGDClassifier(alpha=0.002)        #lsvm performed the best a
model.fit(X_train_count, y_train)
y_pred = model.predict(X_test_count)
print('Lsvm accuracy %s' % accuracy_score(y_pred, y_test))
print('Lsvm precision %s' % precision_score(y_pred, y_test))
print('Lsvm recall %s' % recall_score(y_pred, y_test))
print('Lsvm f1 score %s' % f1_score(y_pred, y_test))
```

```
Lsvm accuracy 0.9148211243611585
Lsvm precision 0.8642384105960265
Lsvm recall 0.9666666666666667
Lsvm f1 score 0.9125874125874126
```

Figure A.6: Training Linear SVM model

A.3.2 Emotion Recognition

Training BiLSTM-BERT model on GoEmotions dataset

```
from tensorflow.keras.layers import Input, Dense, Dropout, Bidirectional, LSTM
from transformers import BertTokenizer, TFBertModel

def create_model(nb_labels):

    tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')

    bert_model = TFBertModel.from_pretrained('bert-base-uncased')

    input_ids = Input(shape=(max_length,), name='input_ids', dtype='int32')
    attention_mask = Input(shape=(max_length,), name='attention_mask', dtype='int32')
    token_ids = Input(shape=(max_length,), name='token_ids', dtype='int32')
    inputs = {'input_ids': input_ids, 'attention_mask': attention_mask, 'token_type_ids': token_ids}

    bert_output = bert_model(inputs)[0]
    lstm_layer = Bidirectional(LSTM(64))(bert_output)
    dropout_layer = Dropout(0.2)(lstm_layer)
    output_layer = Dense(nb_labels, activation='sigmoid', name='output')(dropout_layer)
    model = keras.Model(inputs=inputs, outputs=[output_layer])
    model.compile(loss='binary_crossentropy', optimizer='adam', metrics=['accuracy'])

    return model
```

```
model = create_model(27)

model.summary()
```

Figure A.7: Implementing BiLSTM-BERT

A.3.3 Text Summarization

[14435/14435 1:30:36, Epoch 5/5]

Epoch	Training Loss	Validation Loss	Rouge1	Rouge2	Rougel	Rougelsum	Gen Len
1	3.719000	3.521622	14.553200	2.895700	12.138400	12.743300	17.025500
2	3.666100	3.489677	14.415000	2.859900	12.043500	12.639800	16.749300
3	3.619300	3.478485	14.475500	2.918500	12.168900	12.752400	16.765100
4	3.612400	3.471111	14.515600	2.919500	12.228200	12.811700	16.699100
5	3.569700	3.471188	14.513800	2.916700	12.222700	12.811900	16.707400

Figure A.8: Finetuning T5 model for abstractive text summarization